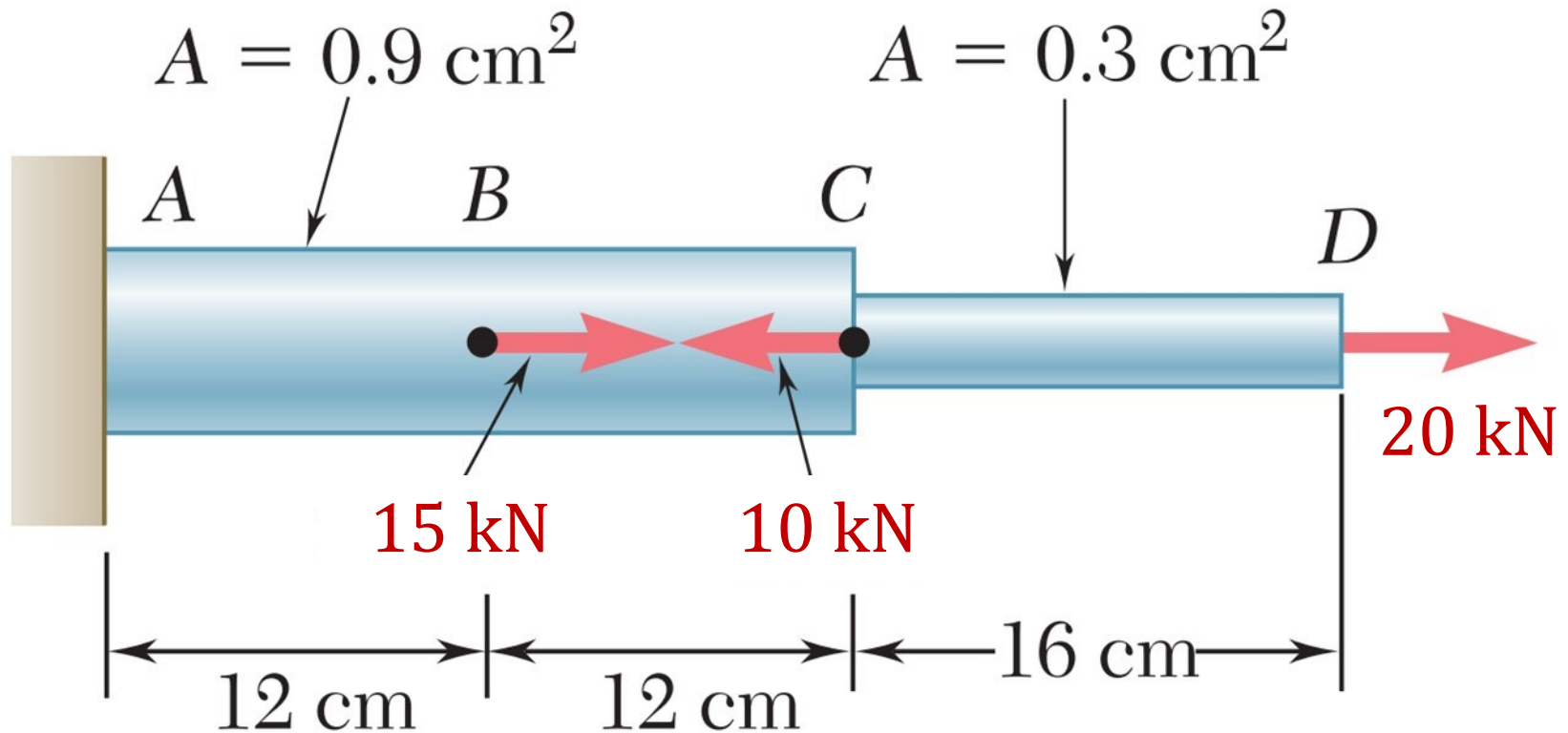


1. Determine the total deformation of the steel rod ( $AD$ ) under the given loads (assuming Young's modulus of both parts  $E = 200 \text{ GPa}$ . The cross-sectional area and the original length of each part is labeled). (10')



# Solutions to Assignment #1:

Since there are different concentrated forces along the axis, the internal resultant force and the stress will not be the same for different regions.

Assuming the  $x$ -axis as shown in the figure, for the right part of section  $aa'$ , the free-body diagram gives the internal resultant force:

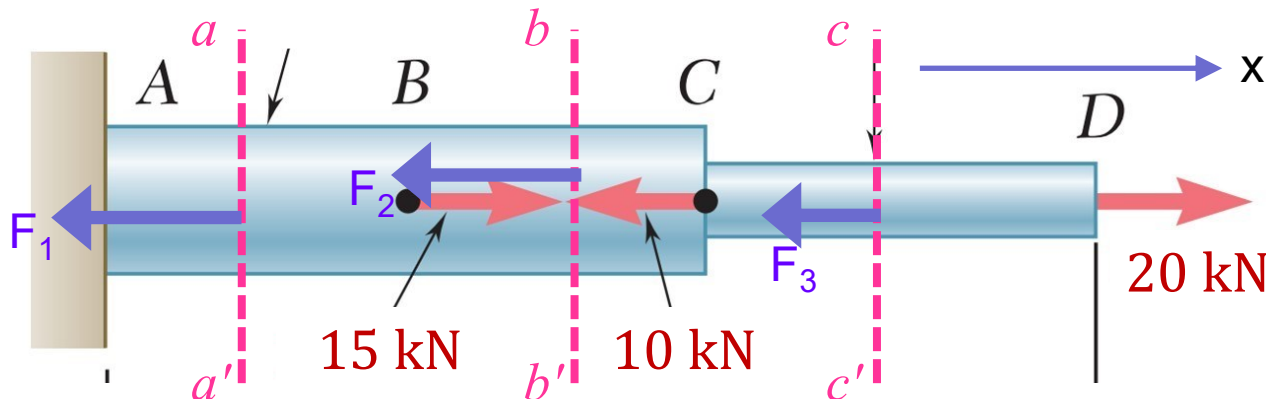
$$F_1 = 20 + 15 - 10 = 25 \text{ kN (tensile)}$$

Stress in region  $AB$  is:

$$\sigma_1 = \frac{F_1}{A_1} = \frac{25000}{0.9 \times 10^{-4}} = 2.7778 \times 10^8 \text{ Pa}$$

→ Using the Hooke's law, the deformation in region  $AB$  is:

$$\delta_{AB} = \frac{\sigma_1 L_{AB}}{E_1} = \frac{2.7778 \times 10^8 \times 0.12}{200 \times 10^9} = 1.6667 \times 10^{-4} \text{ m (tensile)}$$



## Solutions to Assignment #1, cont'd:

For the right part of section bb', the free-body diagram gives the internal resultant force:

$$F_2 = 20 - 10 = 10 \text{ kN (tensile)}$$

Stress in region  $BC$  is:

$$\sigma_2 = \frac{F_2}{A_2} = \frac{10000}{0.9 \times 10^{-4}} = 1.1111 \times 10^8 \text{ Pa}$$

→ Using the Hooke's law, the deformation in region  $BC$  is:

$$\delta_{BC} = \frac{\sigma_2 L_{BC}}{E_2} = \frac{1.1111 \times 10^8 \times 0.12}{200 \times 10^9} = 6.6667 \times 10^{-5} \text{ m (tensile)}$$

For the right part of section cc', the free-body diagram gives the internal resultant force:

$$F_3 = 20 \text{ kN (tensile)}$$

Stress in region  $CD$  is:

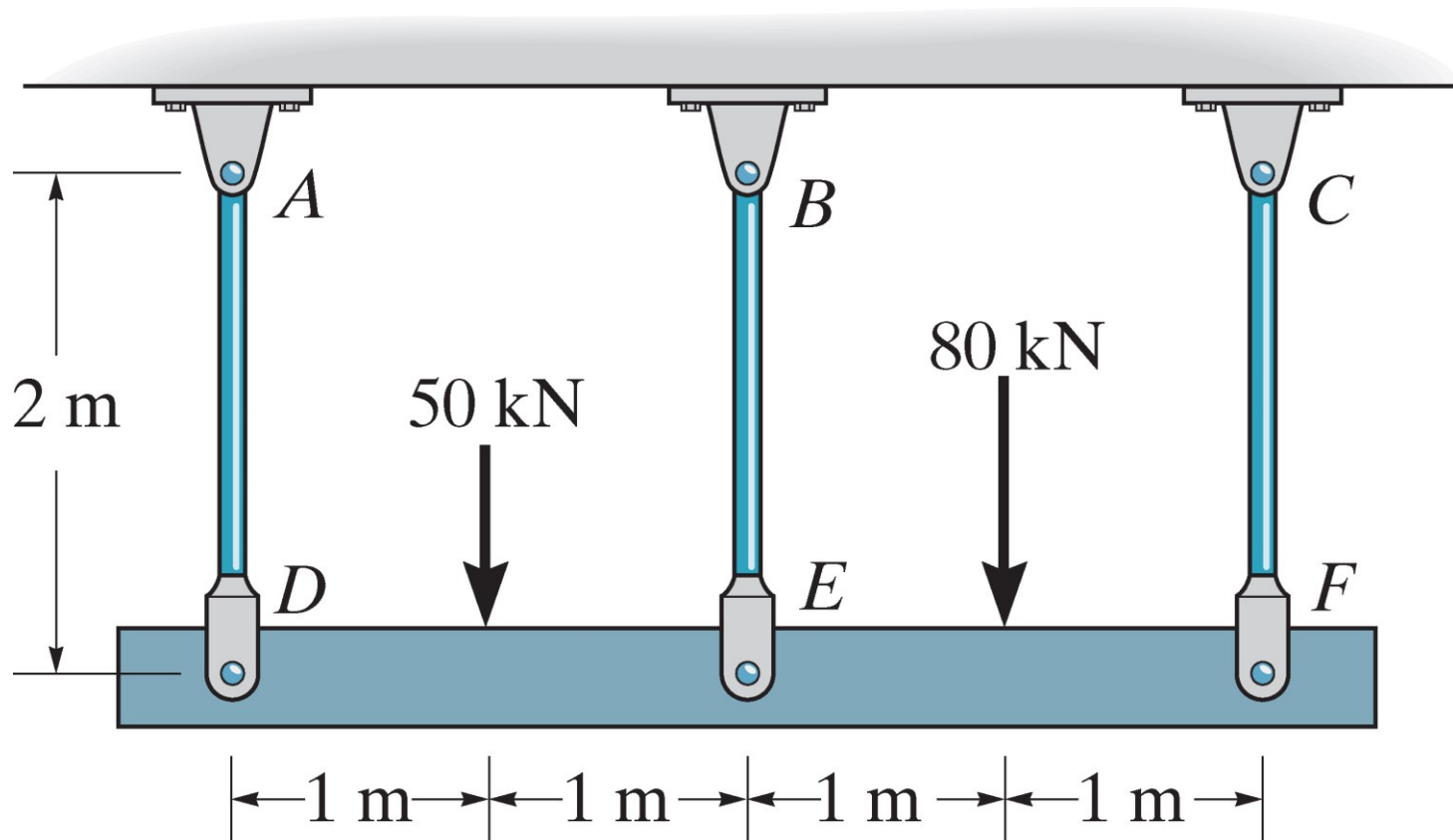
$$\sigma_3 = \frac{F_3}{A_3} = \frac{20000}{0.3 \times 10^{-4}} = 6.6667 \times 10^8 \text{ Pa}$$

→ Using the Hooke's law, the deformation in region  $CD$  is:

$$\delta_{CD} = \frac{\sigma_3 L_{CD}}{E_3} = \frac{6.6667 \times 10^8 \times 0.16}{200 \times 10^9} = 5.3333 \times 10^{-4} \text{ m (tensile)}$$

The total deformation of the rod is:  $\delta_{AD} = \delta_{AB} + \delta_{BC} + \delta_{CD} = 7.6667 \times 10^{-4} \text{ m (Ans.)}$   
(tensile)

2. The three suspender bars are made of same material (modulus of elasticity: 200 GPa) and have equal cross-sectional areas of  $450 \text{ mm}^2$ . Determine the average normal stress in each bar if the rigid beam is subjected to the loading shown. (10')



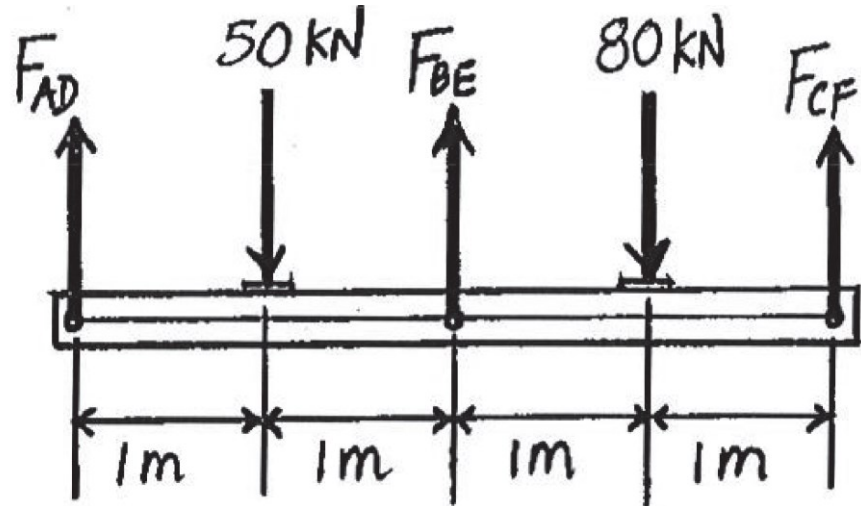
## Solutions to Assignment #2:

### Step 1: Equilibrium

Applying the free-body diagram to the rigid beam.

$$+\uparrow \Sigma F_y = 0; \quad F_{AD} + F_{BE} + F_{CF} - 50(10^3) - 80(10^3) = 0 \quad (1)$$

$$\zeta + \Sigma M_D = 0; \quad F_{BE}(2) + F_{CF}(4) - 50(10^3)(1) - 80(10^3)(3) = 0 \quad (2)$$



## Solutions to Assignment #2, cont'd:

### Step 2 & 3: Compatibility Condition and Load-Displacement Relations

Referring to the geometry shown in the figure:

$$\delta_{BE} = \delta_{AD} + \left( \frac{\delta_{CF} - \delta_{AD}}{4} \right) (2)$$

$$\delta_{BE} = \frac{1}{2} (\delta_{AD} + \delta_{CF})$$

$$\frac{F_{BE} L}{AE} = \frac{1}{2} \left( \frac{F_{AD} L}{AE} + \frac{F_{CF} L}{AE} \right)$$

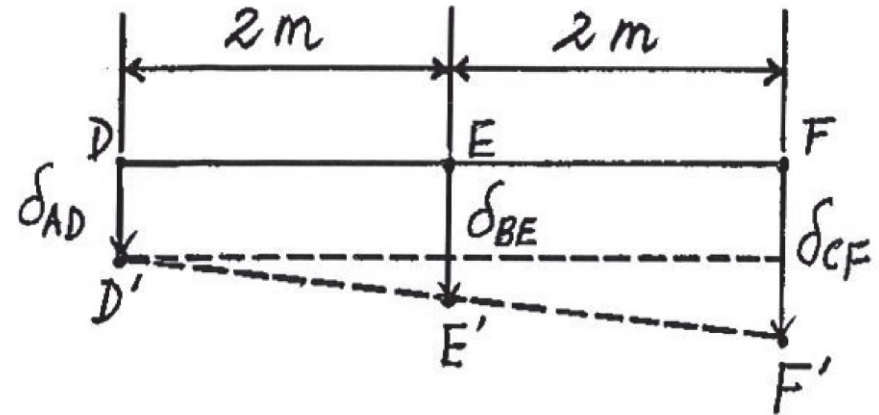
$$F_{AD} + F_{CF} = 2 F_{BE} \quad (3)$$

Solving Eqs. (1), (2), and (3) yields

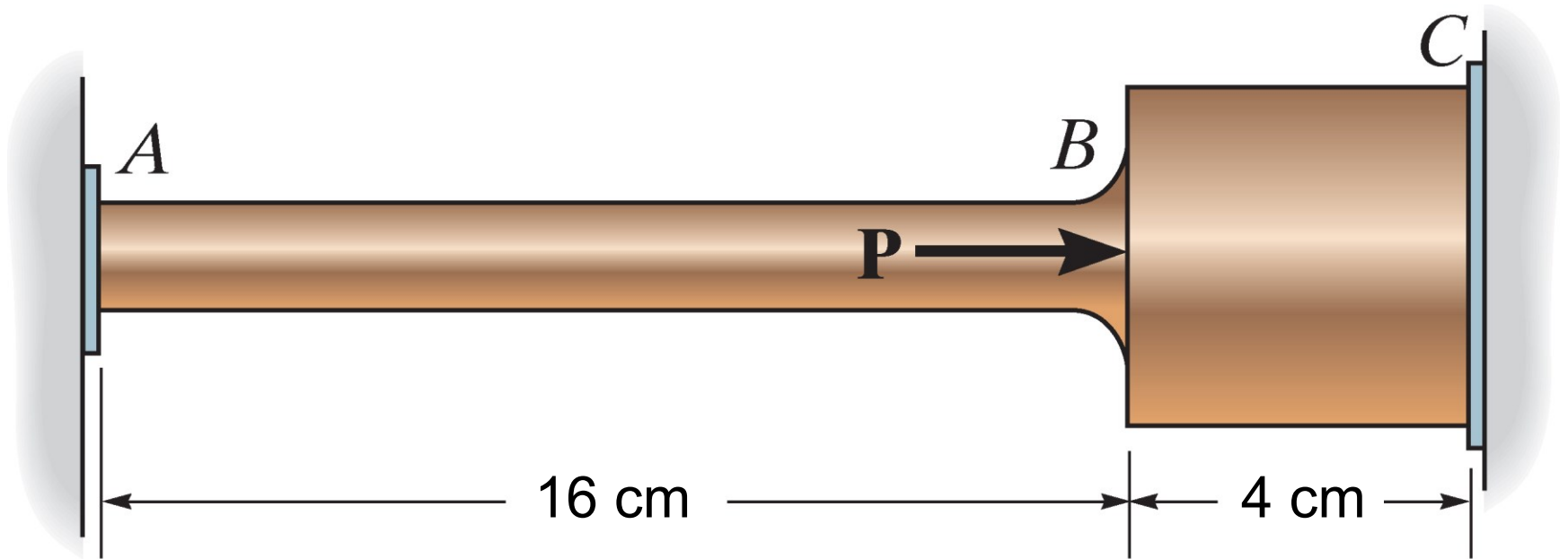
$$F_{BE} = 43.33 \text{ kN}, \quad F_{AD} = 35.83 \text{ kN}, \quad F_{CF} = 50.83 \text{ kN}$$

$$\text{Then, } \sigma_{BE} = \frac{F_{BE}}{A} = \mathbf{96.3 \text{ MPa}} \quad (\text{Ans.})$$

$$\sigma_{AD} = \frac{F_{AD}}{A} = \mathbf{79.6 \text{ MPa}}, \quad \sigma_{CF} = \frac{F_{CF}}{A} = \mathbf{113 \text{ MPa}} \quad (\text{Ans.})$$



3. The force **P** is applied to the section B of the bar, which is made from an idealized elastoplastic material. The bar has cross-sectional areas of  $8 \text{ cm}^2$  in region *AB* and  $24 \text{ cm}^2$  in region *BC*. Take yield strength of the material  $\sigma_Y = 250 \text{ MPa}$ . Determine the maximum force **P** before any part of the bar starts to yield. (15')



## Solutions to Assignment #3:

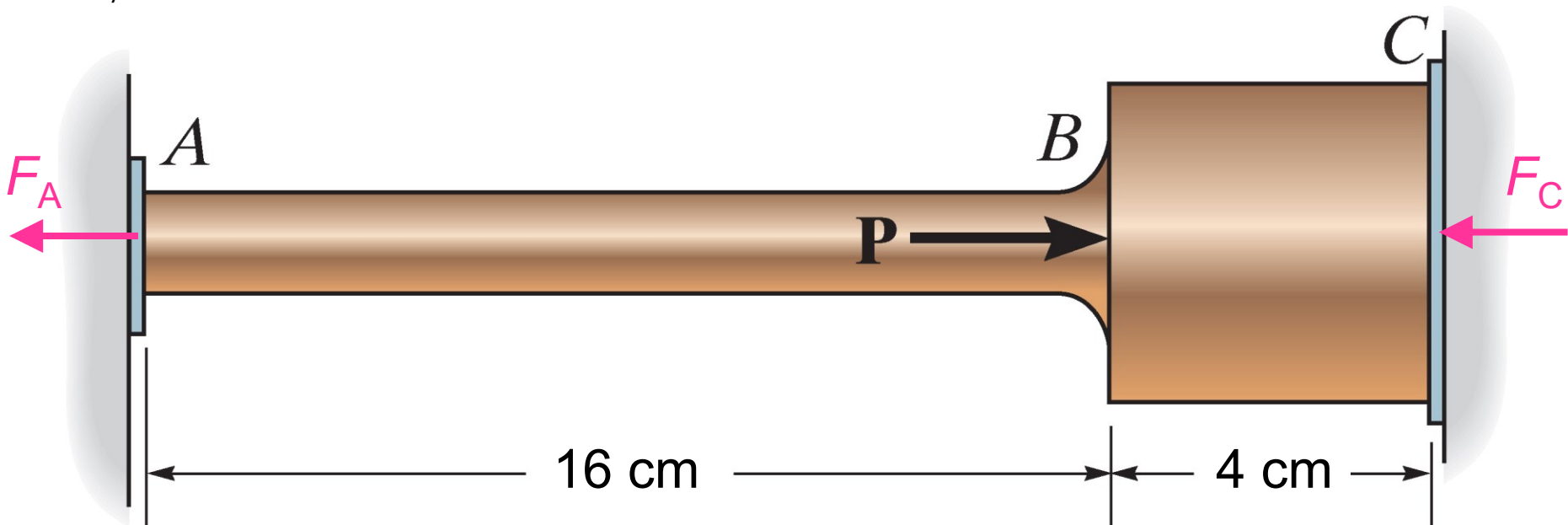
Assuming the resultant reactions at the two ends  $A$  and  $C$  as shown in the figure, for the entire bar, the free-body diagram gives:

$$\sum F_x = 0 : \quad F_A + F_C - P = 0 \quad (1)$$

### Compatibility condition:

The total strain or displacement of the entire bar is zero, i.e.

$$\delta_{A/C} = 0 \rightarrow \delta_{AB} + \delta_{BC} = 0 \quad (2)$$



## Solutions to Assignment #3, cont'd:

The elongation / compression of the bar can be obtained by the force (related to stress) and modulus of elasticity (refer to lecture ppt Ch.4.1 page 7):

$$\delta_{AB} = \frac{F_A L_{AB}}{E_{AB} A_{AB}} \quad \delta_{BC} = - \frac{F_C L_{BC}}{E_{BC} A_{BC}}$$

Substituting these into Eq. (2), we get  $\frac{F_A L_{AB}}{E_{AB} A_{AB}} - \frac{F_C L_{BC}}{E_{BC} A_{BC}} = 0$

$$\rightarrow F_C = 12F_A \quad (3)$$

Substituting (3) into (1), we have  $F_A = \frac{P}{13}, F_C = \frac{12P}{13}$

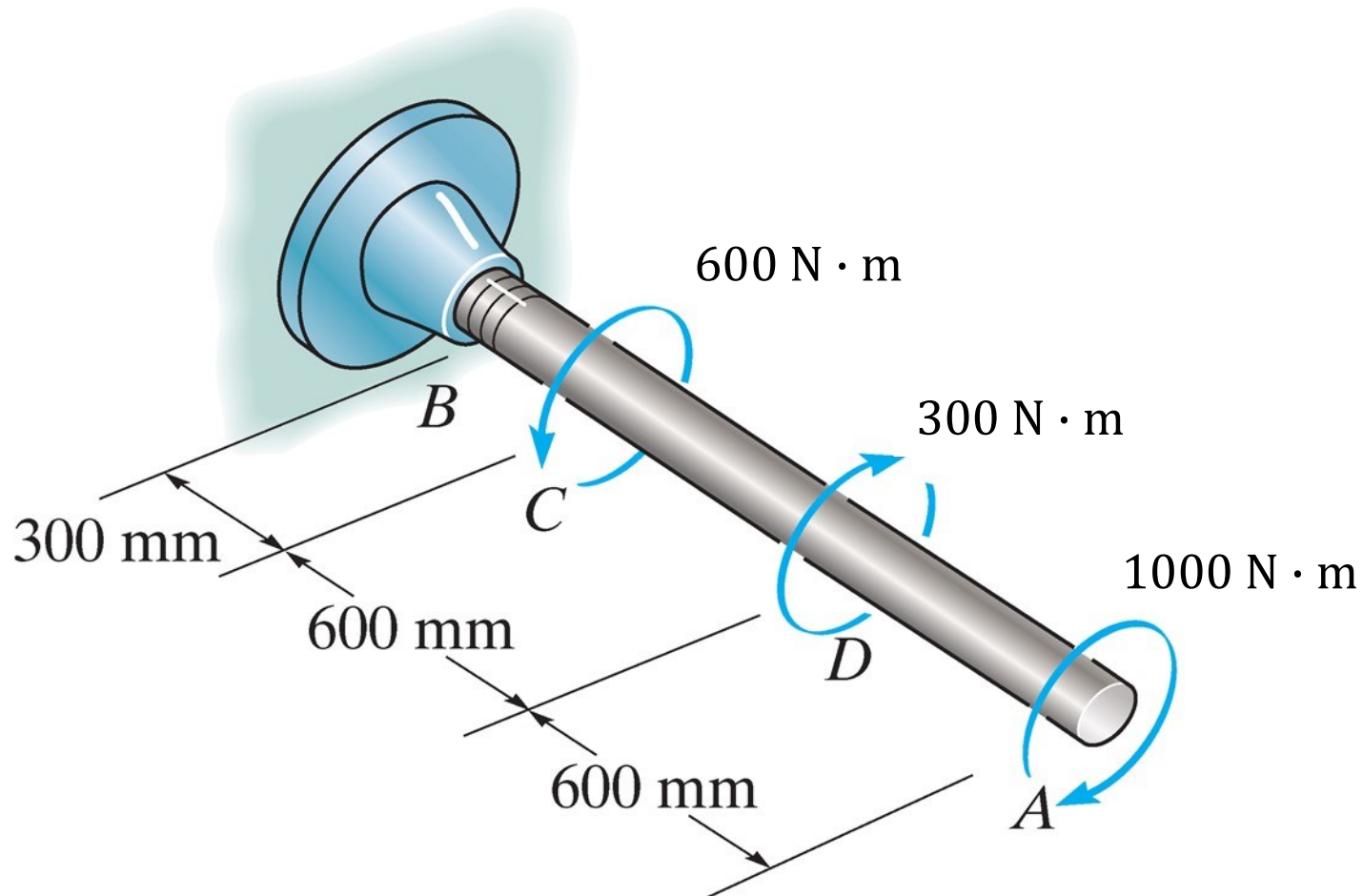
If all bar parts do not yield, the stress everywhere should not exceed the yield strength:

$$\left. \begin{array}{l} \sigma_{AB} = \frac{F_A}{A_{AB}} \leq \sigma_Y \\ \sigma_{BC} = \frac{F_C}{A_{BC}} \leq \sigma_Y \end{array} \right\} \rightarrow \left\{ \begin{array}{l} \frac{P}{13 \times 8 \times 10^{-4}} \leq 250 \times 10^6 \\ \frac{12 \times P}{13 \times 24 \times 10^{-4}} \leq 250 \times 10^6 \end{array} \right. \rightarrow \left\{ \begin{array}{l} P \leq 2600 \text{ kN} \\ P \leq 650 \text{ kN} \end{array} \right.$$

$$\rightarrow P_{\max} = 650 \text{ kN} \quad (\text{Ans.})$$

(the maximum stress happens first in BC)

4. The 50-mm-diameter steel ( $G = 75 \text{ GPa}$ ) shaft is subjected to the torques shown. Determine the maximum shear stress developed in the shaft. Also determine the angle of twist of the end  $A$  with respect to the end  $B$ . (10')



# Solutions to Assignment #4:

## Step 1: Equilibrium

For each segment  $BC$ ,  $CD$ , and  $AD$ , assuming the resultant internal moment as shown in the figure, apply the free-body diagram for each segment (moment equation of equilibrium):

$$\sum M = 0 \rightarrow$$

$$T_{BC} + 600 - 300 - 1000 = 0 \quad (1)$$

$$T_{CD} - 300 - 1000 = 0 \quad (2)$$

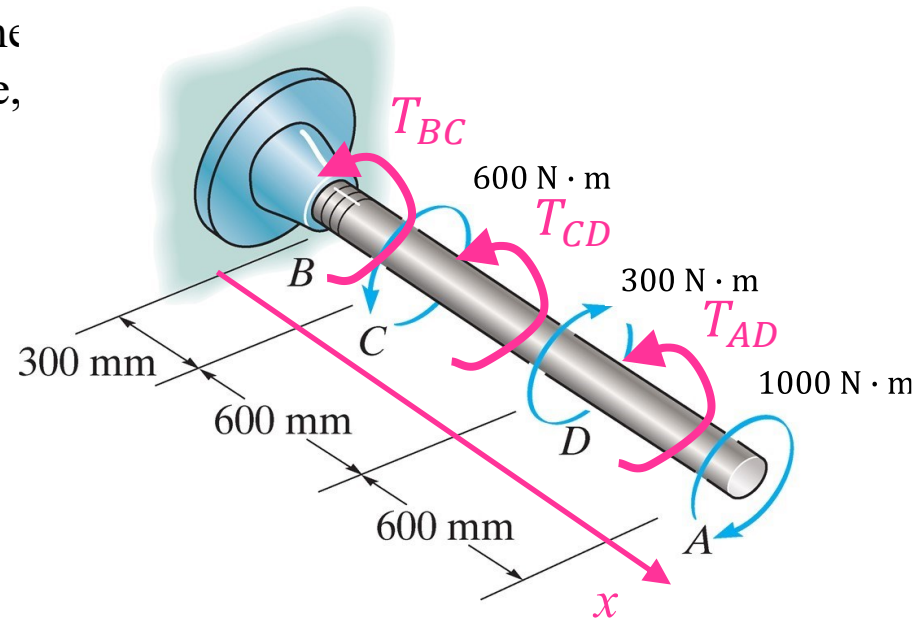
$$T_{AD} - 1000 = 0 \quad (3)$$

Thus,

$$T_{BC} = 700 \text{ N} \cdot \text{m}$$

$$T_{CD} = 1300 \text{ N} \cdot \text{m}$$

$$T_{AD} = 1000 \text{ N} \cdot \text{m}$$



## Solutions to Assignment #4, Cont'd:

### Step 2: Torsion Formula

The maximum shear stress developed in the shaft can be determined by examining the individual part with torsion formula:

$$(\tau_{max})_{BC} = \frac{T_{BC}c}{J} = \frac{700 \times 0.025}{\frac{\pi}{2} \times 0.025^4} = 28.52 \text{ MPa}$$

$$(\tau_{max})_{CD} = \frac{T_{CD}c}{J} = \frac{1300 \times 0.025}{\frac{\pi}{2} \times 0.025^4} = 52.97 \text{ MPa}$$

$$(\tau_{max})_{AD} = \frac{T_{AD}c}{J} = \frac{1000 \times 0.025}{\frac{\pi}{2} \times 0.025^4} = 40.74 \text{ MPa}$$

Then, maximum shear stress developed in the shaft is **52.97 MPa**, which occurs on the surface of region *CD*.     **(Ans.)**

# Solutions to Assignment #4, Cont'd:

## Step 3: Load-Displacement Relations

The angle of twist for each segment of the shaft can be obtained by the torque, shear modulus of elasticity, and the polar moment of inertia:

$$\phi_{BC} = -\frac{T_{BC}L_{BC}}{JG}$$

$$\phi_{CD} = -\frac{T_{CD}L_{CD}}{JG}$$

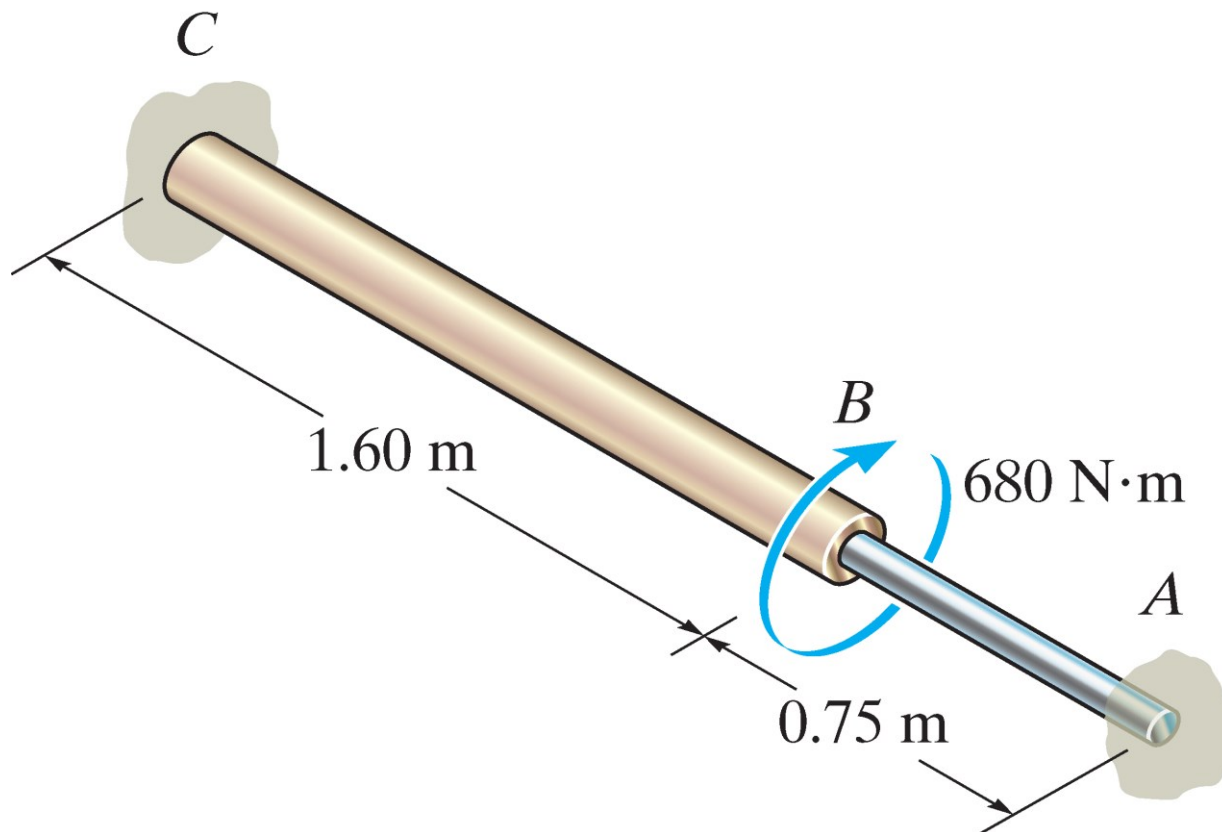
$$\phi_{AD} = -\frac{T_{AD}L_{AD}}{JG}$$

**Note the sign of twist angle!  
(see the right-hand-rule in  
lecture note Ch.5.2 page 8 or  
page 209 in the textbook)**

Then, the total angle of twist is:

$$\begin{aligned}\phi_{A/B} &= \phi_{BC} + \phi_{CD} + \phi_{AD} \\ &= -\frac{700 \times 0.3 + 1300 \times 0.6 + 1000 \times 0.6}{JG} \\ &= -\frac{1590}{\frac{\pi}{2} \times 0.025^4 \times 75 \times 10^9} = \mathbf{-0.03455 \text{ rad}} \quad (\text{Ans.})\end{aligned}$$

5. A rod is made from two segments:  $AB$  is steel and  $BC$  is brass. It is fixed at its ends and subjected to a torque of  $T = 680 \text{ N} \cdot \text{m}$ . If the steel portion has a diameter of 30 mm, determine the required diameter of the brass portion so the reactions at the two ends  $A$  and  $C$  will be the same. Take shear modulus of elasticity of steel and brass as 80 GPa and 40 GPa, respectively. (10')



## Solutions to Problem #5:

### Compatibility Condition:

Since both ends are fixed, the total twist angle of end  $A$  with respect to end  $C$  is zero, i.e.

$$\phi_{A/C} = 0 \rightarrow \phi_{AB} + \phi_{BC} = 0 \quad (1)$$

The angle of twist for each segment of the bar can be obtained by the torque, shear modulus of elasticity, and the polar moment of inertia:

$$\phi_{AB} = \frac{T_A L_{AB}}{J_{AB} G_{AB}} \quad \phi_{BC} = -\frac{T_C L_{BC}}{J_{BC} G_{BC}}$$

Substituting them into Eq. (1), we get

$$\frac{T_A L_{AB}}{J_{AB} G_{AB}} = \frac{T_C L_{BC}}{J_{BC} G_{BC}}$$

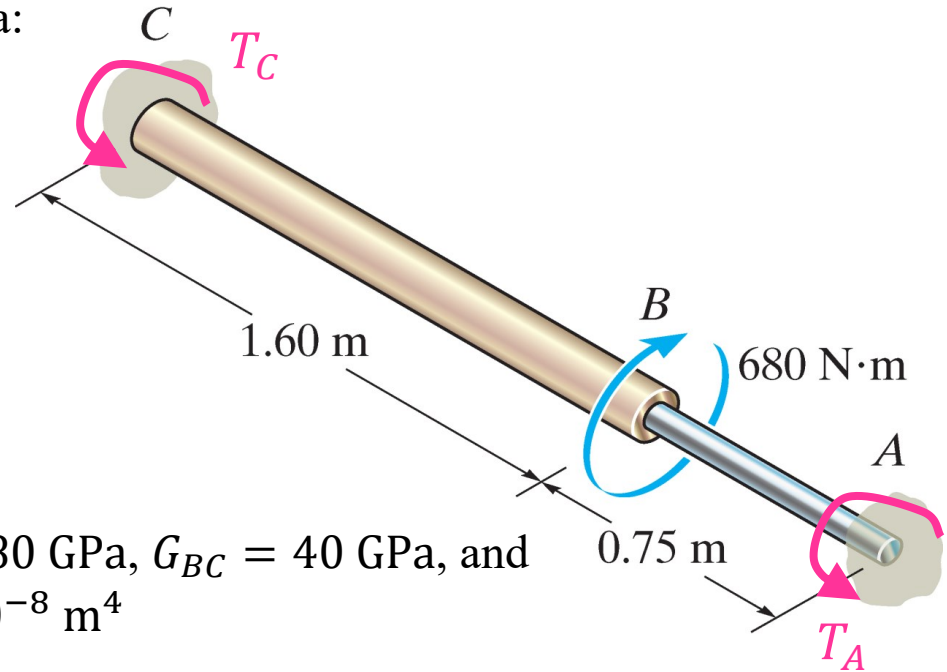
Since  $L_{AB} = 0.75$  m,  $L_{BC} = 1.6$  m,  $G_{AB} = 80$  GPa,  $G_{BC} = 40$  GPa, and

$$J_{AB} = \frac{\pi}{2} \times 0.015^4 = 7.9522 \times 10^{-8} \text{ m}^4$$

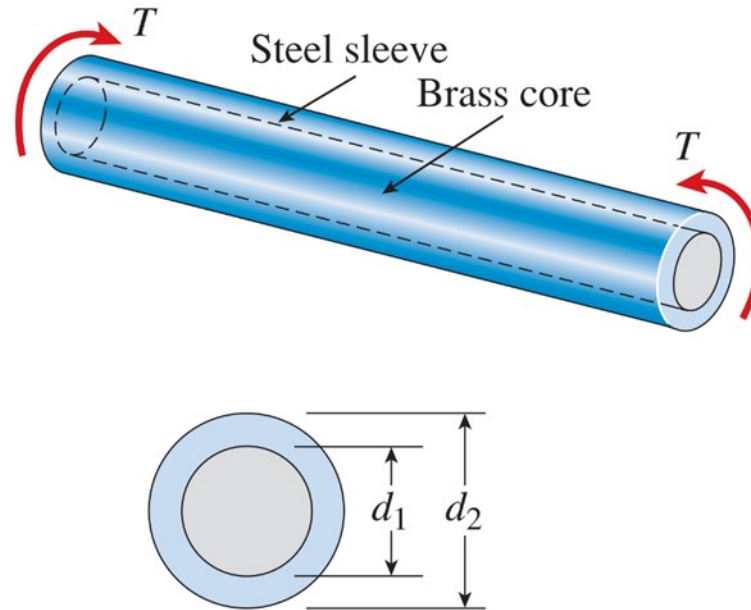
$$J_{BC} = \frac{\pi}{2} \times r_{BC}^4$$

$$T_A = T_C$$

Then,  $r_{BC} = 0.02156$  m, i.e. diameter of brass portion is **43.12 mm (Ans.)**



6. The composite shaft is manufactured by shrink-fitting a steel sleeve over a brass core so that the two parts act as a single solid bar in torsion. The outer diameters of the two parts are  $d_1 = 40$  mm for the brass core and  $d_2 = 48$  mm for the steel sleeve. The shear moduli of elasticity are  $G_{br} = 40$  GPa for the brass and  $G_{st} = 80$  GPa for the steel. Assuming that the allowable shear stresses in the brass and steel are  $\tau_{br} = 50$  MPa and  $\tau_{st} = 80$  MPa, respectively, determine the maximum permissible torque  $T_{max}$  that may be applied to the shaft. (15')



# Solutions to Assignment #6:

## Step 1: Equilibrium

Assuming the reactive torques in the brass core and steel sleeve as  $T_{br}$  and  $T_{st}$ , respectively. For the entire specimen, the free-body diagram (moment equation of equilibrium) gives:

$$\begin{aligned}\sum M &= 0 \rightarrow \\ T_{br} + T_{st} &= T\end{aligned}\quad (1)$$

## Step 2: Compatibility Condition

Since the brass core and steel sleeve are firmly connected, the angle of twist for brass core and steel sleeve should be equal, i.e.

$$\phi_{br} = \phi_{st} \quad (2)$$

## Step 3: Load-Displacement Relations

Using the angle of twist for each component of the composite shaft we have:

$$\begin{aligned}\phi_{br} &= \frac{T_{br}L}{J_{br}G_{br}} \\ \phi_{st} &= \frac{T_{st}L}{J_{st}G_{st}}\end{aligned}$$

## Solutions to Assignment #6, cont'd:

Substituting them into Eq. (2) and considering  $J_{\text{br}} = \frac{\pi}{2} \times 0.02^4$ ,  $J_{\text{st}} = \frac{\pi}{2} \times (0.024^4 - 0.02^4)$ ,  $G_{\text{br}} = 40 \text{ GPa}$ ,  $G_{\text{st}} = 80 \text{ GPa}$ , we get

$$T_{\text{st}} = 2.1472T_{\text{br}} \quad (3)$$

Combining Eqs. (1) and (3) we obtain

$$T_{\text{br}} = 0.3177T \quad T_{\text{st}} = 0.6823T$$

### Step 4: Maximum Shearing Stress

The maximum shear stress in the brass core

$$(\tau_{\text{max}})_{\text{br}} = \frac{T_{\text{br}}c_1}{J_{\text{br}}} = \frac{0.3177T \times 0.02}{\frac{\pi}{2} \times 0.02^4} \leq \tau_{\text{br}|_{\text{limit}}} = 50 \text{ MPa}$$

$$\rightarrow T \leq 1977.71 \text{ N} \cdot \text{m}$$

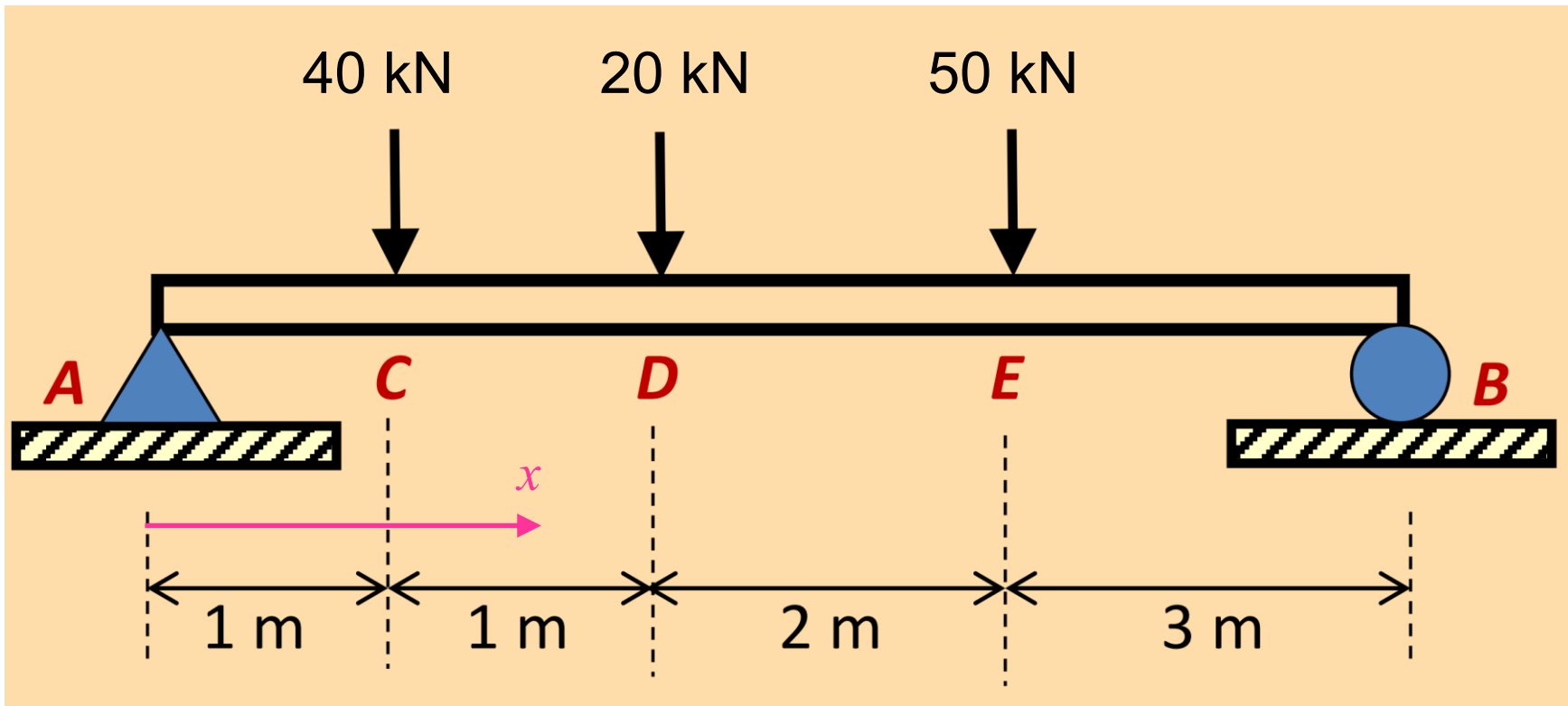
The maximum shear stress in the steel sleeve

$$(\tau_{\text{max}})_{\text{st}} = \frac{T_{\text{st}}c_2}{J_{\text{st}}} = \frac{0.6823T \times 0.024}{\frac{\pi}{2} \times (0.024^4 - 0.02^4)} \leq \tau_{\text{st}|_{\text{limit}}} = 80 \text{ MPa}$$

$$\rightarrow T \leq 1318.21 \text{ N} \cdot \text{m}$$

Therefore, the maximum permissible torque  $T_{\text{max}} = \mathbf{1318.21 \text{ N} \cdot \text{m.}}$  (Ans.)

7. Using the section method and equilibrium equations, determine the shear and moment for the beam as a function of position  $x$  and plot the diagrams. Reaction forces can occur in both horizontal and vertical directions at joint  $A$ , while only vertical force can occur at joint  $B$ . (15')



# Solutions to Assignment #7:

## Step 1: Support Reactions

Applying free-body diagram to the entire beam:

$$\sum F_y = 0 \rightarrow$$

$$F_A + F_B = 40 + 20 + 50 = 110 \text{ kN}$$

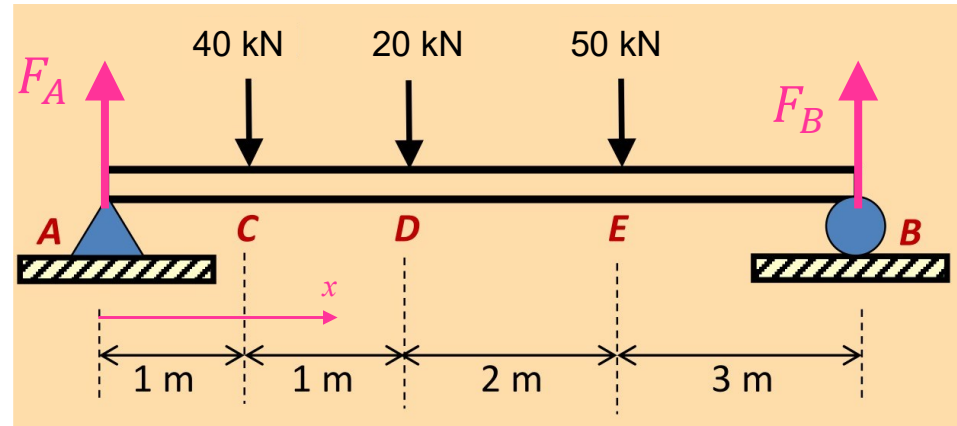
$$\sum M_A = 0 \rightarrow$$

$$F_B \times 7 = 40 \times 1 + 20 \times 2 + 50 \times 4$$

$\rightarrow$

$$F_A = 70 \text{ kN}$$

$$F_B = 40 \text{ kN}$$



# Solutions to Assignment #7, Cont'd:

## Step 2: Shear and Moment Functions

Since the external loading has discontinuous change at point C/D/E, we need to split the beam into four regions (*AC*, *CD*, *DE*, and *EB*).

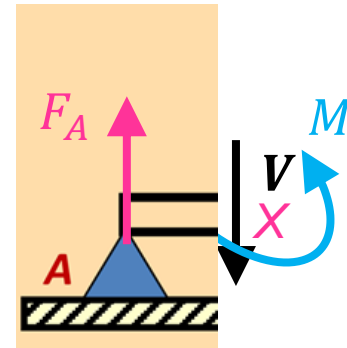
For region *AC*, taking a small segment starting at point *A* with arbitrary length of  $x$  (section position), apply the free-body diagram to the segment:

$$\sum F_y = 0 \rightarrow$$

$$V = F_A = 70 \text{ kN} \quad (0 \leq x \leq 1 \text{ m})$$

$$\sum M_X = 0 \rightarrow M = F_A \cdot x$$

$$\rightarrow M = 70x \quad (0 \leq x \leq 1 \text{ m})$$



# Solutions to Assignment #7, Cont'd:

## Step 2: Shear and Moment Functions, Cont'd

For region CD:

$$\sum F_y = 0 \rightarrow$$

$$V = F_A - 40 = 30 \text{ kN} \quad (1 \leq x \leq 2 \text{ m})$$

$$\sum M_X = 0 \rightarrow M + 40 \cdot (x - 1) = F_A \cdot x$$

$$\rightarrow M = 30x + 40 \text{ kN} \cdot \text{m} \quad (1 \leq x \leq 2 \text{ m})$$

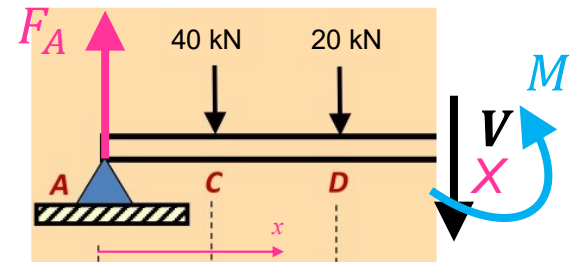
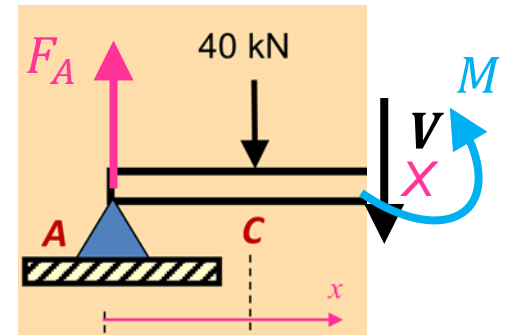
For region DE:

$$\sum F_y = 0 \rightarrow$$

$$V = F_A - 60 = 10 \text{ kN} \quad (2 \leq x \leq 4 \text{ m})$$

$$\sum M_X = 0 \rightarrow M + 40 \cdot (x - 1) + 20 \cdot (x - 2) = F_A \cdot x$$

$$\rightarrow M = 10x + 80 \text{ kN} \cdot \text{m} \quad (2 \leq x \leq 4 \text{ m})$$



## Solutions to Assignment #7, Cont'd:

### Step 2: Shear and Moment Functions, Cont'd

For region EB:

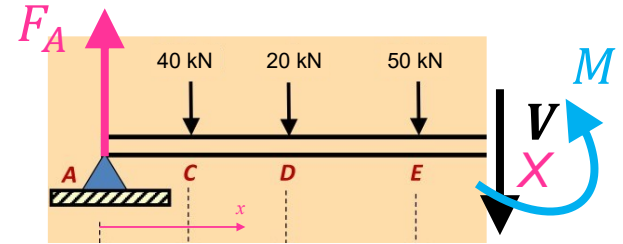
$$\sum F_y = 0 \rightarrow$$

$$V = F_A - 40 - 20 - 50 = -40 \text{ kN} \quad (4 \leq x \leq 7 \text{ m})$$

$$\sum M_x = 0 \rightarrow$$

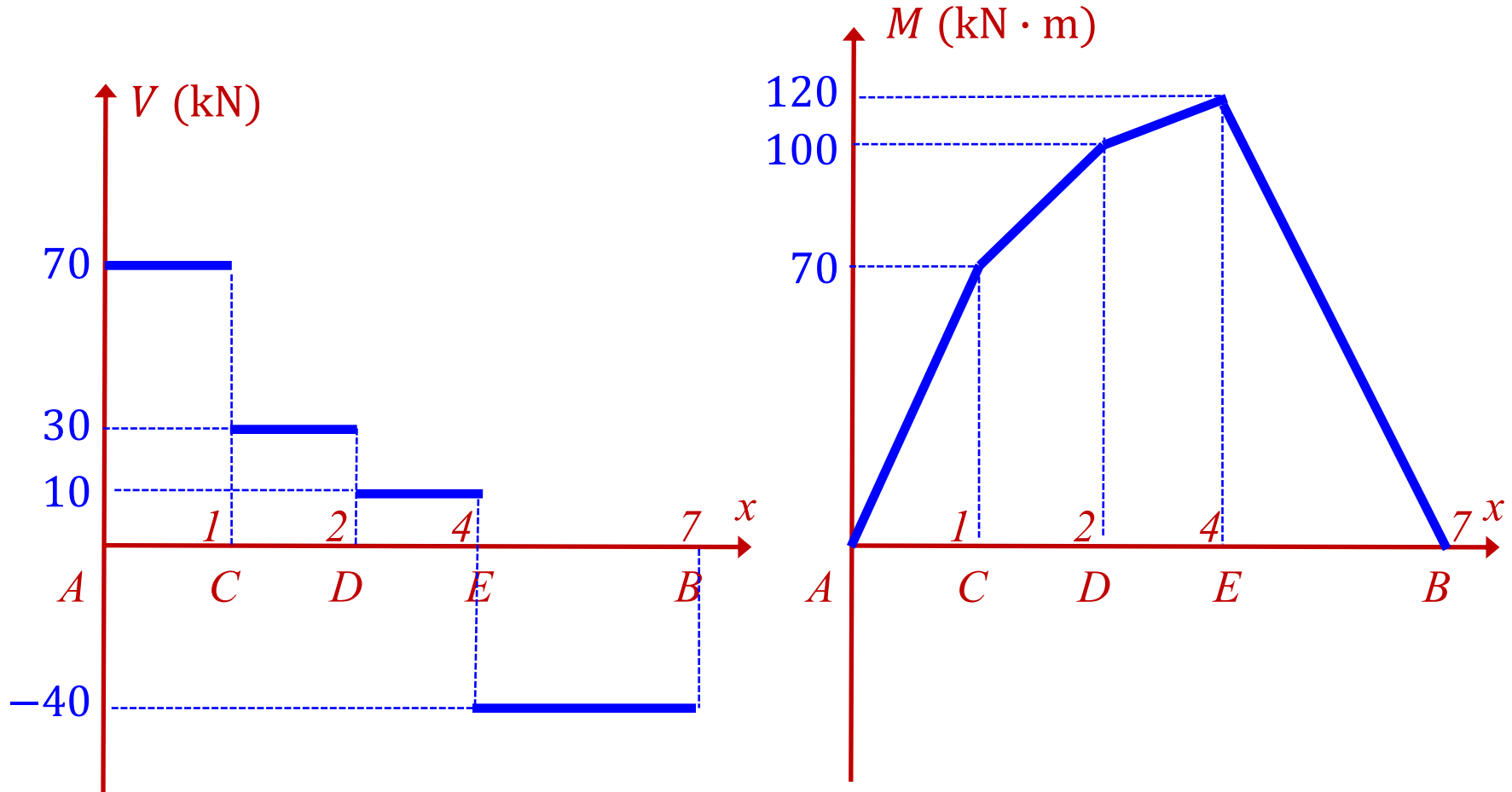
$$M + 40 \cdot (x - 1) + 20 \cdot (x - 2) + 50 \cdot (x - 4) = F_A \cdot x$$

$$\rightarrow \quad M = -40x + 280 \text{ kN} \cdot \text{m} \quad (4 \leq x \leq 7 \text{ m})$$

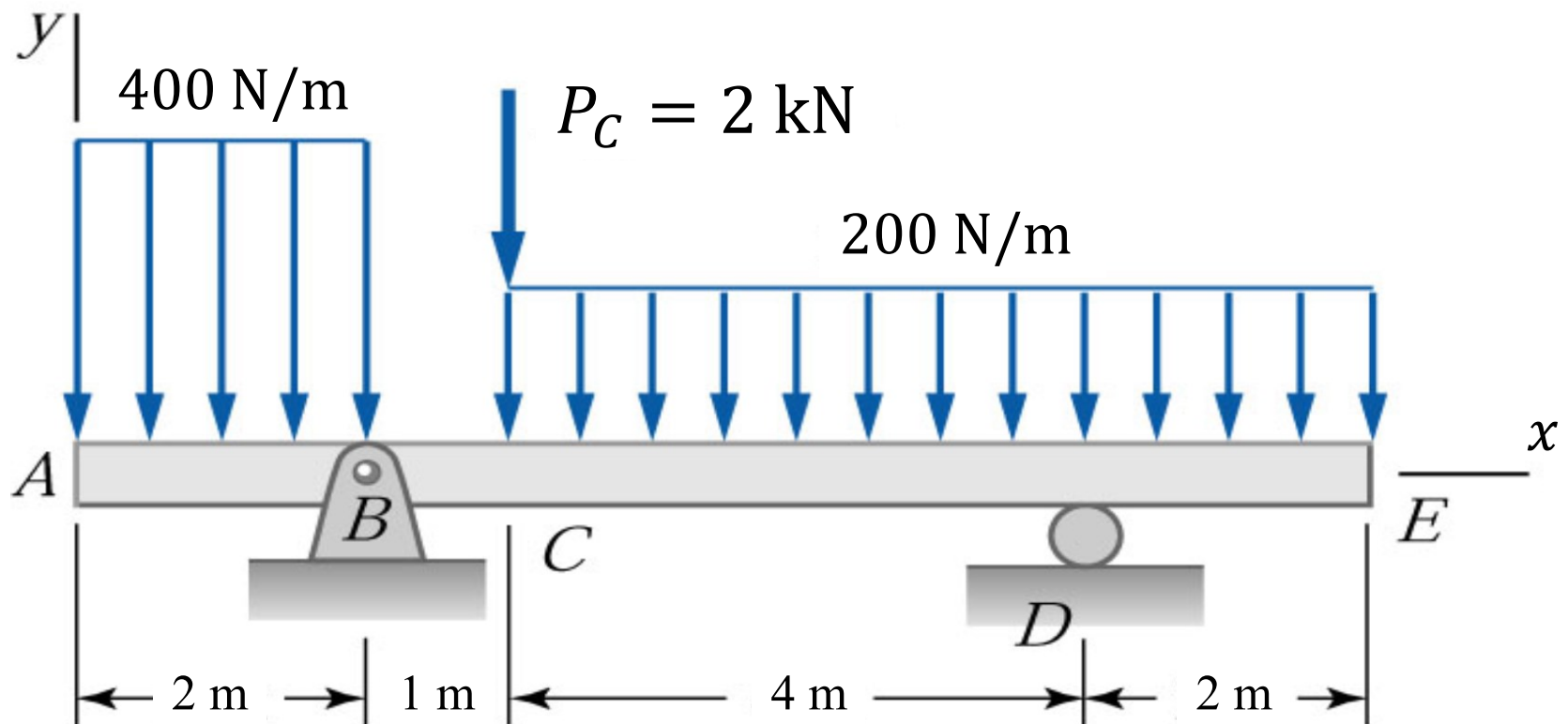


# Solutions to Assignment #7, Cont'd:

## Step3: Shear and Moment Diagrams



8. Determine the absolute maximum normal stress in the beam with external loading shown below. The beam has a uniform square cross-section with lateral size  $a = 0.1$  m. Note: the shear and moment diagrams can be calculated either by section method or graphical method. (15')



# Solutions to Assignment #8:

## Step 1: Support Reactions

Applying free-body diagram to the entire beam:

$$\sum F_y = 0 \rightarrow$$

$$F_B + F_D = 400 \times 2 + 200 \times 6 + 2000 \\ = 4000$$

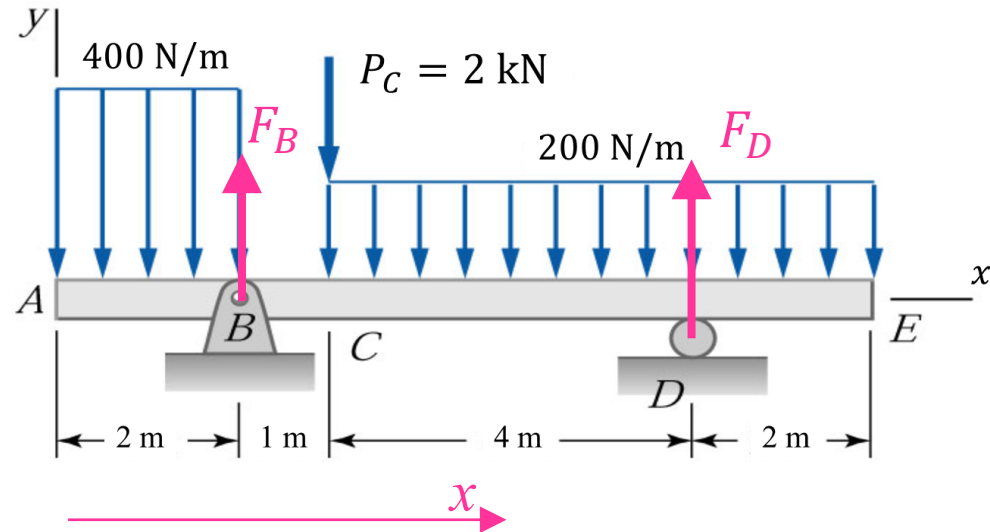
$$\sum M_B = 0 \rightarrow$$

$$F_D \times 5 + 400 \times 2 \times 1 = 200 \times 6 \times 4 \\ + 2000 \times 1$$

$\rightarrow$

$$F_B = 2800 \text{ N}$$

$$F_D = 1200 \text{ N}$$



## Solutions to Assignment #8, cont'd:

### Step 2: Shear Diagram

a) Determine the shear at left end first (the origin  $A$  of  $x$ -axis):

$$x = 0, \quad V = 0$$

b) Determine  $V$  diagram according to  $\frac{dV}{dx} = w$  for  $\underline{AB}$  region:

$$\frac{dV}{dx} = w = -400$$

$$\rightarrow V = -400x + C_1$$

The constant  $C_1$  can be determined by applying the “**boundary**” condition ( $x = 0$ ,  $V = 0$ ). Thus,  $C_1 = 0$  and  $V = -400x$  ( $0 \leq x \leq 2$ ).

c) At point  $B$ , due to the supporting reaction  $F_B = 2800$  N (upward), the  $V$  increases (upward jump) by 2800 N. Then, the  $V$  jumps from  $-800$  to  $+2000$  N. See the shear diagram for details.

## Solutions to Assignment #8, cont'd:

### Step 2: Shear Diagram, cont'd

d) Determine  $V$  diagram according to  $\frac{dV}{dx} = w$  for  $\underline{BC}$  region:

$$\frac{dV}{dx} = w = 0$$

$$\rightarrow V = C_2$$

The constant  $C_2$  can be determined by the “**continuity**” condition at last point  $B$  after considering the jump ( $x = 2$ ,  $V = 2000$ ). Thus,  $C_2 = 2000$  and  $V = 2000$  ( $2 \leq x \leq 3$ ).

e) At point  $C$ , due to the external concentrated force  $P_C = 2000$  N (downward), the  $V$  decreases (downward jump) by 2000 N. Then, the  $V$  jumps from +2000 to 0. See the shear diagram for details.

## Solutions to Assignment #8, cont'd:

### Step 2: Shear Diagram, cont'd

f) Determine  $V$  diagram according to  $\frac{dV}{dx} = w$  for CD region:

$$\frac{dV}{dx} = w = -200$$

$$\rightarrow V = -200x + C_3$$

The constant  $C_3$  can be determined by the “**continuity**” condition at last point  $C$  after considering the jump ( $x = 3$ ,  $V = 0$ ). Thus,  $C_3 = 600$  and  $V = -200x + 600$  ( $3 \leq x \leq 7$ ).

g) At point  $D$ , due to the supporting reaction  $F_D = 1200$  N (upward), the  $V$  increases (upward jump) by 1200 N. Then, the  $V$  jumps from  $-800$  N to  $+400$  N. See the shear diagram for details.

## Solutions to Assignment #8, cont'd:

### Step 2: Shear Diagram, cont'd

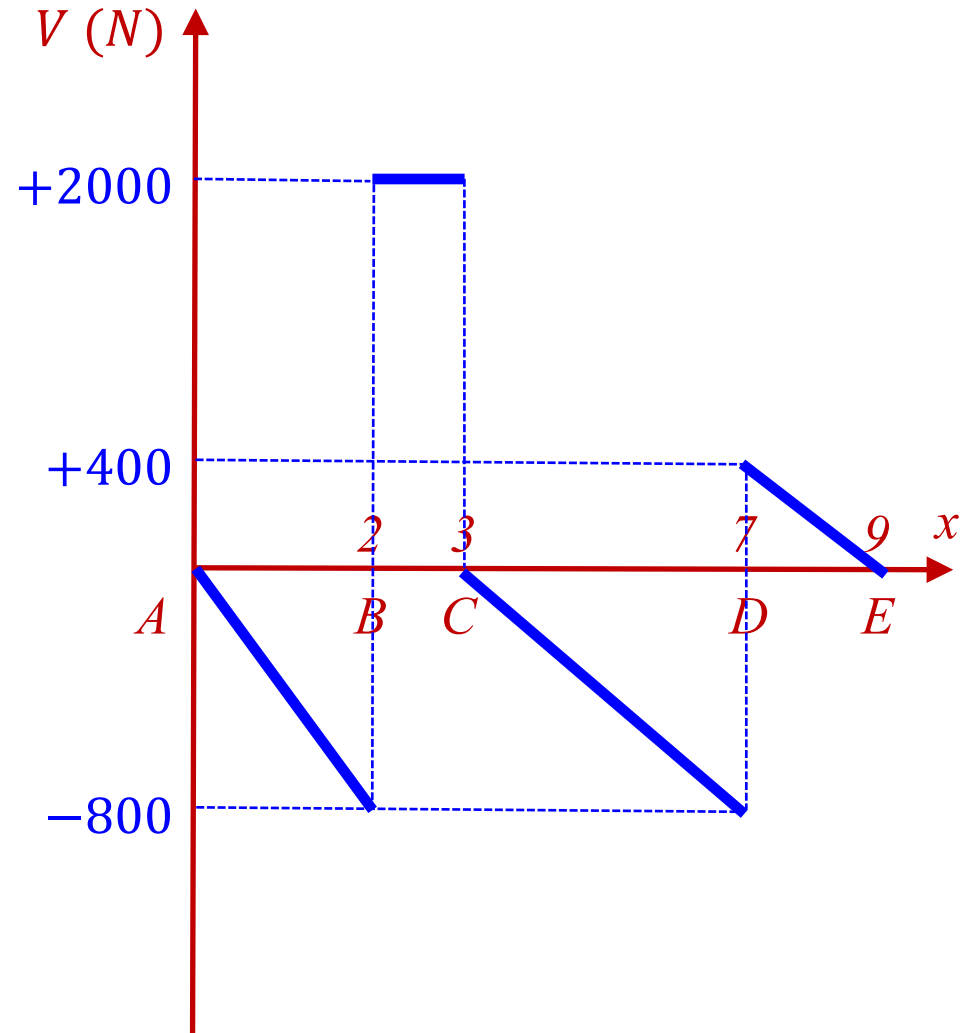
h) Determine  $V$  diagram according to  $\frac{dV}{dx} = w$  for  $\underline{DE}$  region:

$$\frac{dV}{dx} = w = -200$$

$$\rightarrow V = -200x + C_4$$

The constant  $C_4$  can be determined by the “**continuity**” condition at last point  $D$  after considering the jump ( $x = 7$ ,  $V = 400$ ). Thus,  $C_4 = 1800$  and  $V = -200x + 1800$  ( $7 \leq x \leq 9$ ).

The shear diagram is plotted as shown.



## Solutions to Assignment #8, cont'd:

### Step 3: Moment Diagram

a) Determine the moment at left end first (the origin  $A$  of  $x$ -axis):

$$x = 0, \quad M = 0$$

b) Determine **M** diagram according to  $\frac{dM}{dx} = V$  for AB region:

$$\begin{aligned} \frac{dM}{dx} &= V = -400x \\ \rightarrow M &= -200x^2 + C_5 \end{aligned}$$

The constant  $C_5$  can be determined by applying the “**boundary**” condition ( $x = 0$ ,  $M = 0$ ). Thus,  $C_5 = 0$  and  $M = -200x^2$  ( $0 \leq x \leq 2$ ).

c) Determine **M** diagram according to  $\frac{dM}{dx} = V$  for BC region:

$$\begin{aligned} \frac{dM}{dx} &= V = 2000 \\ \rightarrow M &= 2000x + C_6 \end{aligned}$$

The constant  $C_6$  can be determined by applying the “**continuity**” condition at last point  $B$  ( $x = 2$ ,  $M = -800$ ). Thus,  $C_6 = -4800$  and  $M = 2000x - 4800$  ( $2 \leq x \leq 3$ ).

## Solutions to Assignment #8, cont'd:

### Step 3: Moment Diagram, cont'd

d) Determine **M** diagram according to  $\frac{dM}{dx} = V$  for CD region:

$$\frac{dM}{dx} = V = -200x + 600$$

$$\rightarrow M = -100x^2 + 600x + C_7$$

The constant  $C_7$  can be determined by the “**continuity**” condition at last point  $C$  ( $x = 3$ ,  $M = 1200$ ). Thus,  $C_7 = 300$  and  $M = -100x^2 + 600x + 300$  ( $3 \leq x \leq 7$ ).

e) Determine **M** diagram according to  $\frac{dM}{dx} = V$  for DE region:

$$\frac{dM}{dx} = V = -200x + 1800$$

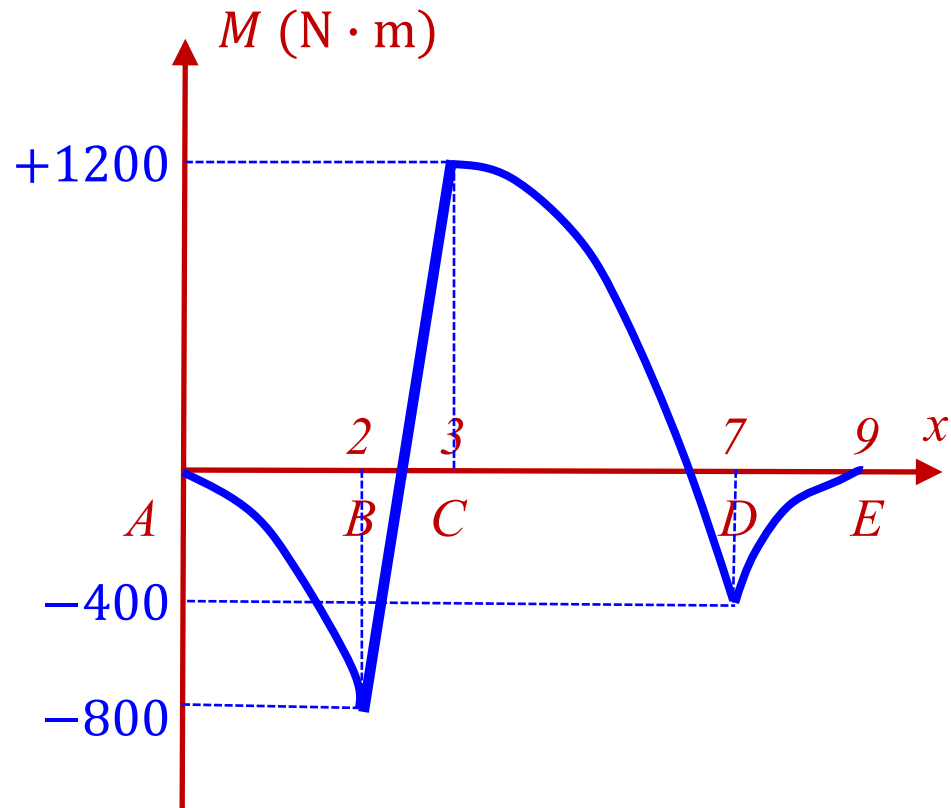
$$\rightarrow M = -100x^2 + 1800x + C_8$$

The constant  $C_8$  can be determined by the “**continuity**” condition at last point  $D$  ( $x = 7$ ,  $M = -400$ ). Thus,  $C_8 = -8100$  and  $M = -100x^2 + 1800x - 8100$  ( $7 \leq x \leq 9$ ).

## Solutions to Assignment #8, cont'd:

### Step 3: Moment Diagram, cont'd

The moment diagram is plotted as shown. The maximum bending moment occurs at  $x = 3$  m, and the corresponding absolute value of moment is  $1200 \text{ N}\cdot\text{m}$ .



## Solutions to Assignment #8, cont'd:

### Step 3: Section Property

The moment of inertia  $I$  of sectional area about the neutral axis is (refer to the inside cover of the textbook for rectangular cross-section):

$$I = \frac{1}{12}bh^3 = \frac{1}{12} \times 0.1 \times 0.1^3 = 8.3333 \times 10^{-6} \text{ m}^4$$

### Step 4: Maximum Normal Stress

The maximum normal stress is determined by:

$$\sigma_{\max} = \frac{Mc}{I}$$

$$\text{Thus, } \sigma_{\max} = \frac{1200 \times 0.1 / 2}{8.3333 \times 10^{-6}} = \mathbf{7.2 \text{ MPa}} \quad (\text{Ans.})$$